

# ISAD FINAL PRESENTATION

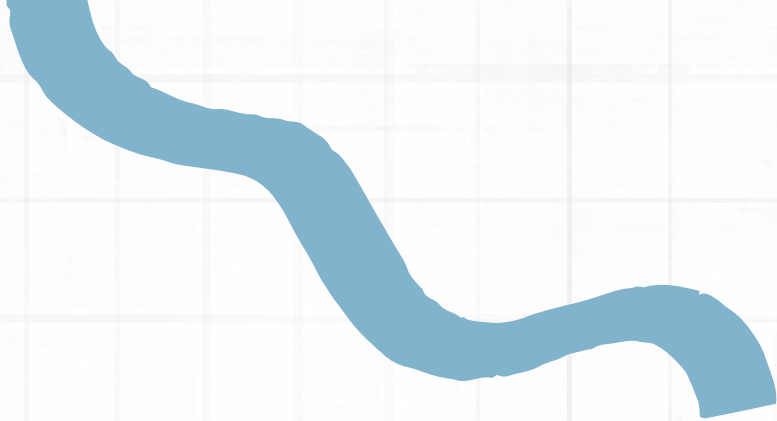
## LINGUAL LEAP

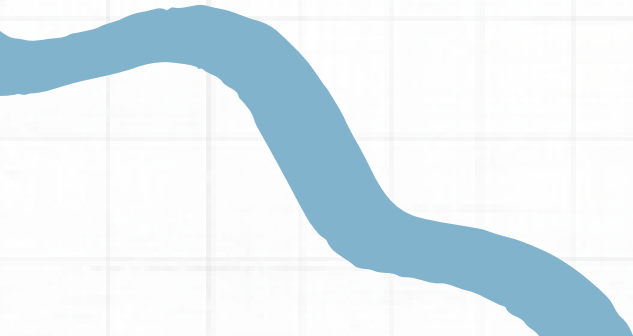


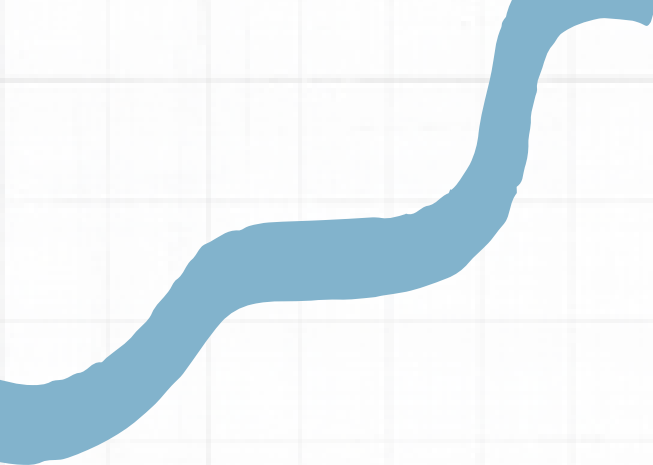
LANGUAGE TRANSLATOR APP

SHRUTHI SURESH  
SAMHITA KOMANDUR SREENIVASA  
PADMA RISHITHA PUSAPATI  
ANIRUDH BATHULA NARSIMHA

# ABSTRACT



- 1. Addressing Language Barriers:** Develop a comprehensive translation app to overcome challenges in international travel, business, and communication.
  - 2. Real-Time Capabilities:** Implement real-time translation for spoken and written input to facilitate seamless communication.
  - 3. Offline Functionality:** Ensure accessibility without an internet connection using locally stored language data.
  - 4. Versatile Features:** Include text-to-speech, speech-to-text, document translation, and interactive language learning exercises.
  - 5. Broad Application:** Support tourists, businesses, and educational users, promoting global connectivity and cultural exchange.
- 



# PART 1:

## 1.1 TRIGGER POINT

### Opportunity: Translate Seamlessly

#### **1.Globalization Impact:**

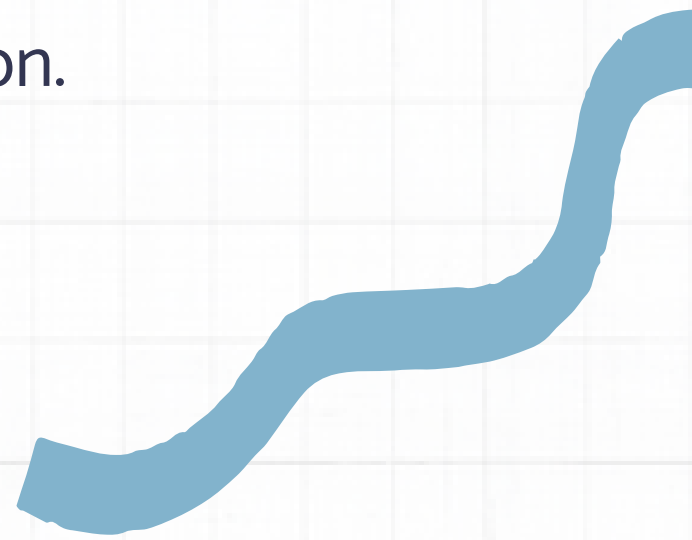
Increasing globalization demands effective cross-language communication.

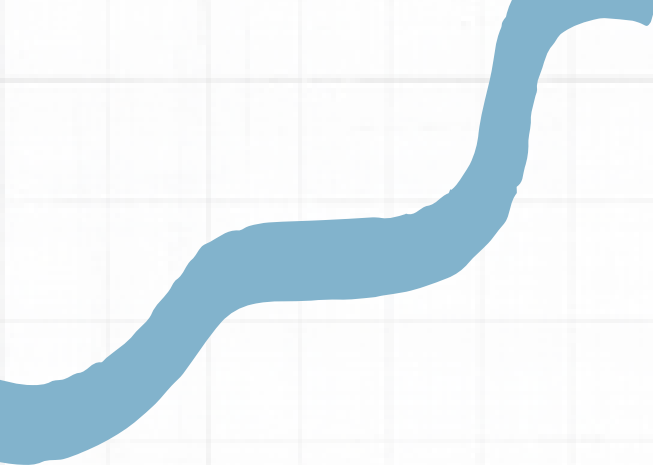
#### **2..Common Challenges:**

Language barriers hinder travel, business, and social interactions.

#### **3.Technological Solution:**

Develop an app with advanced translation technologies to facilitate seamless communication.



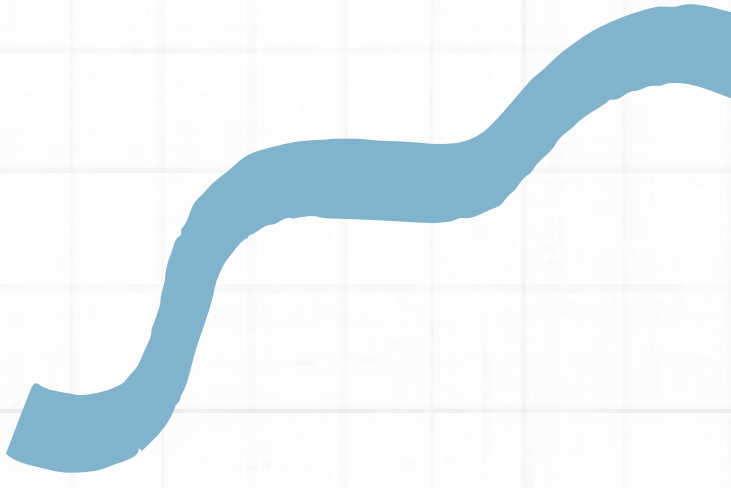


# PART 1.2 :GENERATE SYSTEM VISION DOCUMENT

## **System Capabilities:**

1. Real-Time Translation
2. Offline Mode
3. Text-to-Speech
4. Speech-to-Text
5. Document Translation
6. Language Learning
7. Real-Time Interpretation
8. Educational Translation
9. Business Support
10. Tourist Assistance

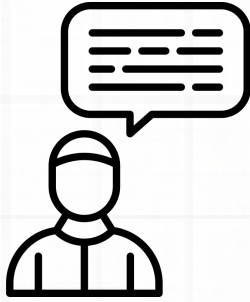
## **Business Benefits:**

1. Enhanced Travel Communication
  2. Cultural Exchange Promotion
  3. Global Market Access
  4. Reduced Misunderstandings
  5. Increased Productivity
- 

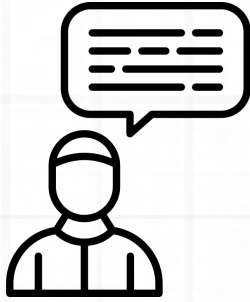


# PART 2

## USER STORIES

|                                                                                     |                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | As a <b>frequent traveler</b> , I want to be able to communicate with locals in their native language in real-time, so that I can easily ask for directions, order food, and engage in conversations without language barriers.                                |
|   | As a <b>business professional</b> , I need the ability to translate important documents and emails into multiple languages offline, so that I can effectively communicate with international clients and partners even when I'm not connected to the internet. |
|  | As a <b>language learner</b> , I want to practice my listening and speaking skills by using the app's text-to-speech and speech-to-text features to convert written texts and spoken words between different languages.                                        |
|  | As a <b>tourist</b> , I want to explore foreign cities confidently by using the app's real-time translation feature to understand street signs, menus, and public transportation announcements in languages I'm not familiar with.                             |

## 2.1 USER STORIES

|                                                                                     |                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | As a <b>parent</b> , I want my children to have access to educational content in multiple languages, so that they can learn about different cultures and languages from an early age using the app's translation and language learning features. |
|   | As a <b>conference attendee</b> , I need to be able to understand presentations and engage with international speakers during Q&A sessions by using the app's real-time translation feature for spoken language.                                 |
|  | As a <b>customer service representative</b> , I want to provide support to international customers by using the app's translation capabilities to communicate with them in their preferred language via chat and email.                          |
|  | As a <b>journalist</b> , I need to conduct interviews with people from diverse linguistic backgrounds, so I rely on the app's speech-to-text feature to transcribe spoken interviews into written text in different languages.                   |

## 2.1 USER STORIES



As a **language teacher**, I want to create interactive lessons and exercises for my students using the app's text-to-speech and speech-to-text features to provide pronunciation practice and feedback.



As a **remote worker**, I need to collaborate with colleagues from around the world, so I use the app's translation capabilities to communicate effectively in team meetings, brainstorming sessions, and project discussions.





# USE CASES

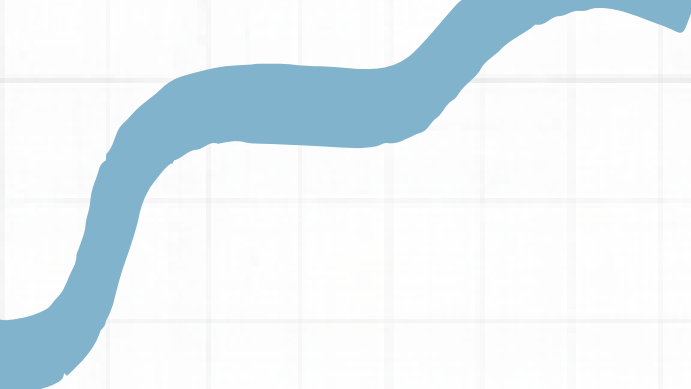
|                                                                 |                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Real-Time Language Translation</b>                        | The user initiates real-time translation by speaking into the app's microphone or typing text. The app translates the input into the desired language and provides the translated text or spoken response instantly.                                              |
| <b>2. Offline Translation Mode</b>                              | In scenarios where internet connectivity is limited or unavailable, the user activates offline translation mode. The app accesses locally stored language data to perform translations, ensuring uninterrupted communication even without an internet connection. |
| <b>3. Text-to-Speech Conversion</b>                             | The user inputs written text into the app, such as a document or message, and selects the desired language. The app converts the text into spoken words, allowing the user to listen to the content in their preferred language.                                  |
| <b>4. Interpreting for Deaf and Hard of Hearing Individuals</b> | Sign language interpreters use the app to communicate with deaf and hard of hearing individuals who may not share the same spoken language, ensuring effective communication and access to information.                                                           |
| <b>5. Language Identification</b>                               | The user captures or uploads audio or text content in an unknown language. The app identifies the language and provides options for translation or further actions based on the detected language.                                                                |



# USE CASES

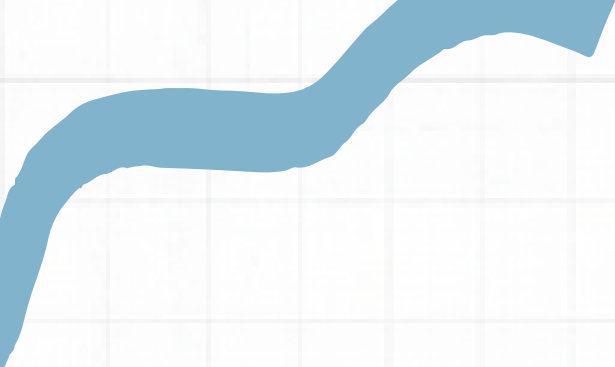
|                                             |                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6. Interpretation in Social Settings</b> | In scenarios where internet connectivity is limited or unavailable, the user activates offline translation mode. The app accesses locally stored language data to perform translations, ensuring uninterrupted communication even without an internet connection.                                |
| <b>7. Business Document Translation</b>     | The user interacts with the app to translate business documents, emails, or presentations into different languages to ensure clear and accurate communication in professional settings.                                                                                                          |
| <b>8. Tourist Assistance</b>                | The user navigates through unfamiliar locations while traveling. They utilize the app to translate menus, or directions into their native language, enhancing their ability to engage with local culture.                                                                                        |
| <b>9. Language Learning Exercises</b>       | The user accesses language learning exercises within the app, such as vocabulary quizzes or pronunciation drills. They interact with the exercises by listening to spoken words, matching translations, or recording their own voice for pronunciation practice.                                 |
| <b>10. Language Tutoring and Feedback</b>   | The user engages in language tutoring sessions within the app, practicing speaking, listening, or writing skills. The app provides feedback on pronunciation, grammar, and vocabulary usage, helping the user improve their language proficiency through personalized exercises and assessments. |





# USE CASES

|                                                 |                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>11. Multilingual Social Networking</b>       | The user interacts with others on a social networking platform within the app. The app supports multilingual communication, allowing users to post, comment, and engage with content in their preferred language while seamlessly translating interactions with users who speak different languages. |
| <b>12. Cross-Cultural Business Negotiations</b> | A business executive uses the app to bridge language gaps during international negotiations, ensuring mutual understanding and successful deal-making.                                                                                                                                               |
| <b>13. Medical Interpretation</b>               | A healthcare provider communicates with patients from diverse linguistic backgrounds using the app to ensure accurate diagnosis and treatment recommendations.                                                                                                                                       |
| <b>14. Legal Interpreting</b>                   | A lawyer utilizes the app to communicate with clients and witnesses who speak different languages during legal proceedings, ensuring effective representation and comprehension of legal matters.                                                                                                    |
| <b>15. Emergency Response</b>                   | Emergency responders use the app to communicate with individuals in distress who speak different languages, ensuring prompt and accurate assistance during emergencies.                                                                                                                              |



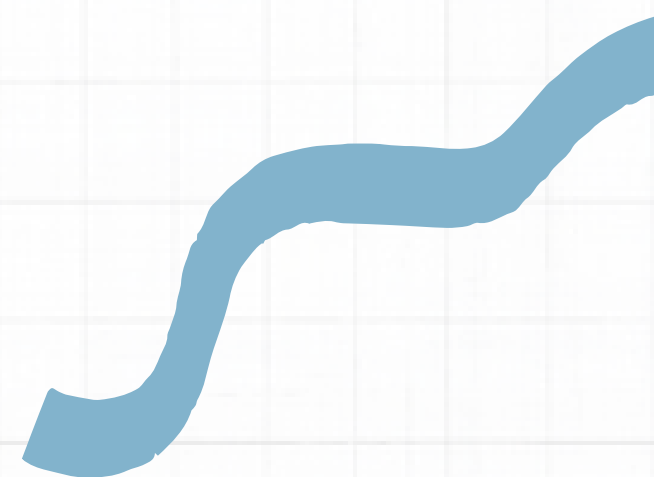
# USE CASES

## **16. Cultural Sensitivity Training**

HR professionals use the app to deliver cultural sensitivity training to employees, helping them understand and respect cultural differences in the workplace.

## **17. Language Access in Government Services**

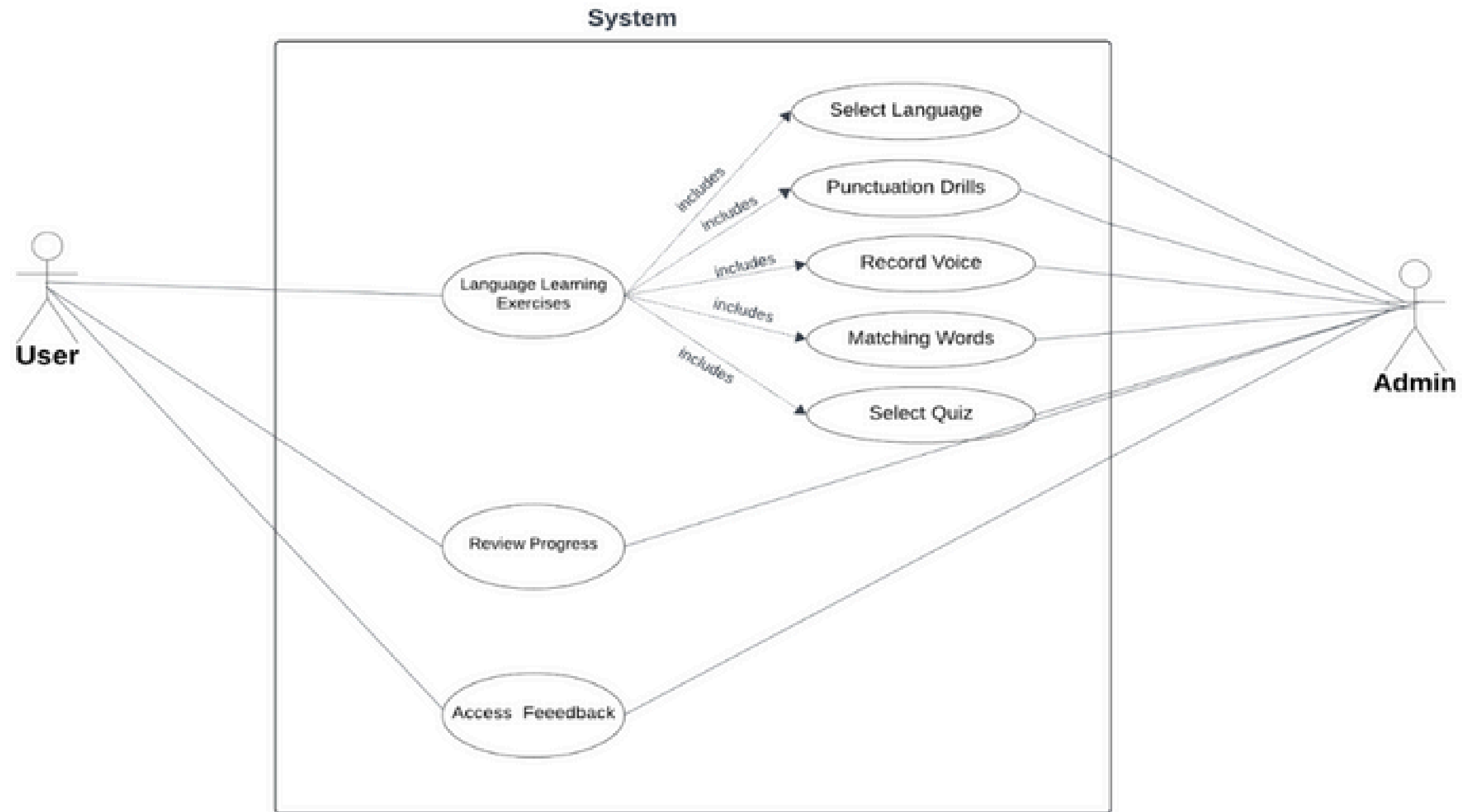
Government agencies use the app to provide language access services to citizens with limited English proficiency, ensuring equitable access to government programs and services.



# 1ST FULLY DEVELOPED USE CASE DESCRIPTION

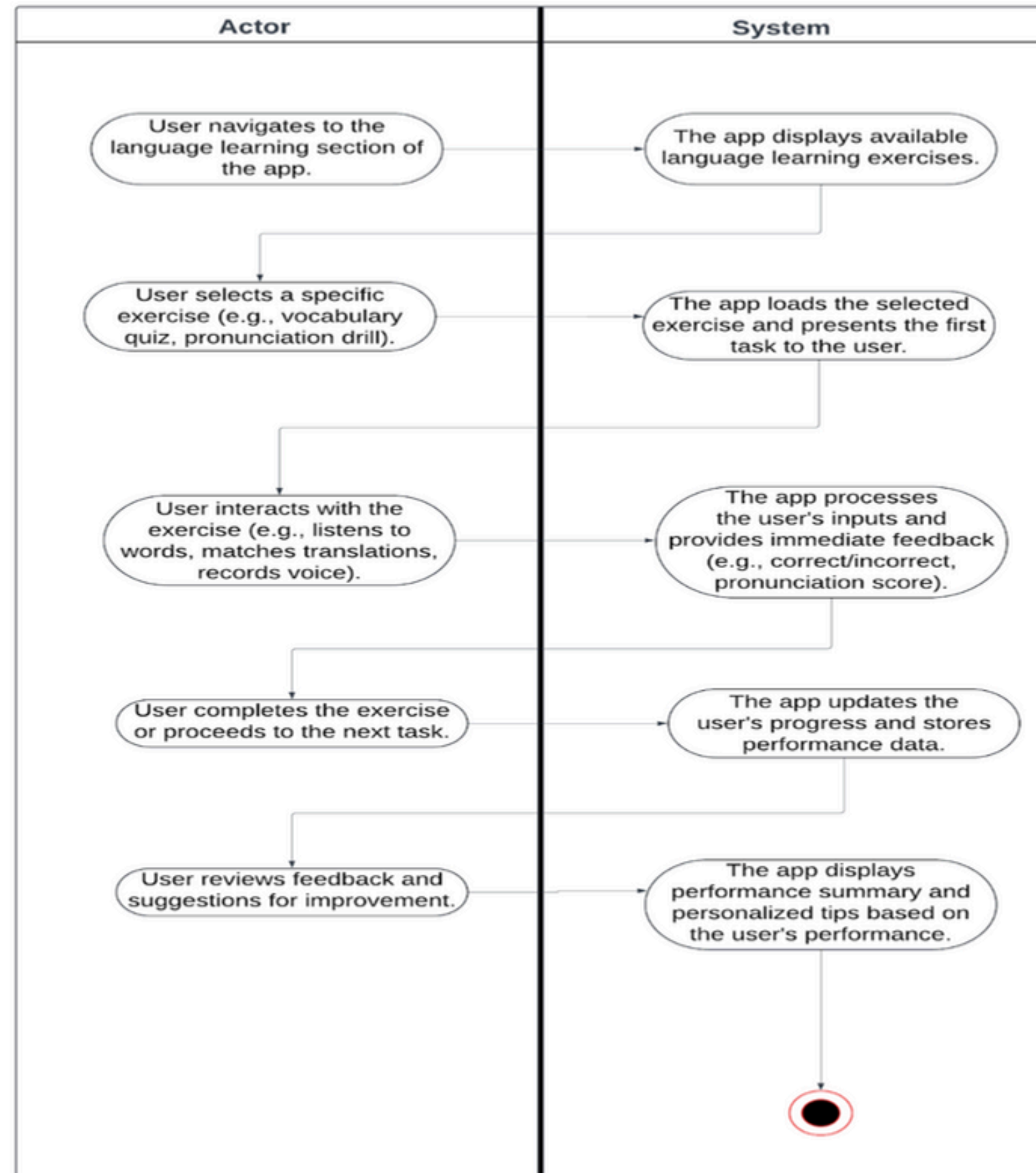
| Use case name        | Language Learning Exercises                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scenario             | The user accesses language learning exercises within the app, such as vocabulary quizzes or pronunciation drills.                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Triggering event     | The user selects the language learning section in the app.                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Brief description    | The user engages in various language learning exercises, interacting with the app to listen to spoken words, match translations, or record their own voice for pronunciation practice.                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Actors               | User, System(app)                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Related use cases    | Real-Time Language Translation, Speech-to-Text Conversion                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Stakeholders         | Users                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Preconditions        | The user is logged into the app. The app has language learning content available and functioning.                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Postconditions       | The user completes language learning exercises and receives feedback. The user's progress is recorded and updated.                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Flow of activities   | Actor                                                                                                                                                                                                                                                                                                                                                                                  | System                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                      | 1. User navigates to the language learning section of the app.<br>2. User selects a specific exercise (e.g., vocabulary quiz, pronunciation drill).<br>3. User interacts with the exercise (e.g., listens to words, matches translations, records voice).<br>4. User completes the exercise or proceeds to the next task.<br>5. User reviews feedback and suggestions for improvement. | 1.1 The app displays available language learning exercises.<br>2.1 The app loads the selected exercise and presents the first task to the user.<br>3.1 The app processes the user's inputs and provides immediate feedback (e.g., correct/incorrect, pronunciation score).<br>4.1 The app updates the user's progress and stores performance data.<br>5.1 The app displays performance summary and personalized tips based on the user's performance. |
| Exception conditions | 1. If the app fails to load an exercise, it should display an error message and suggest troubleshooting steps (e.g., check internet connection, restart app).<br>2. If the user encounters a technical issue (e.g., microphone not working), the app should provide guidance on how to resolve the issue.                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

# USECASE DIAGRAM

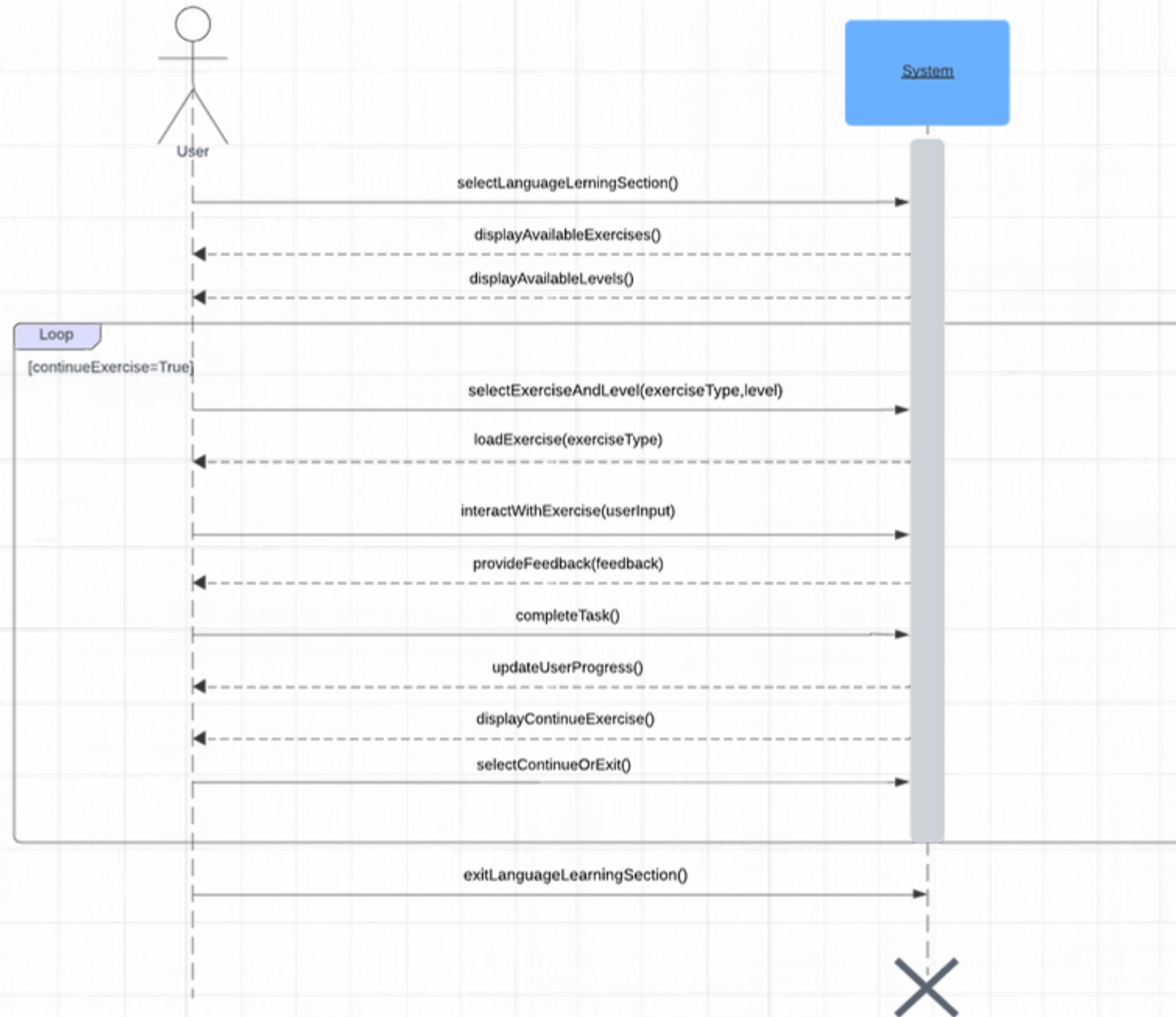




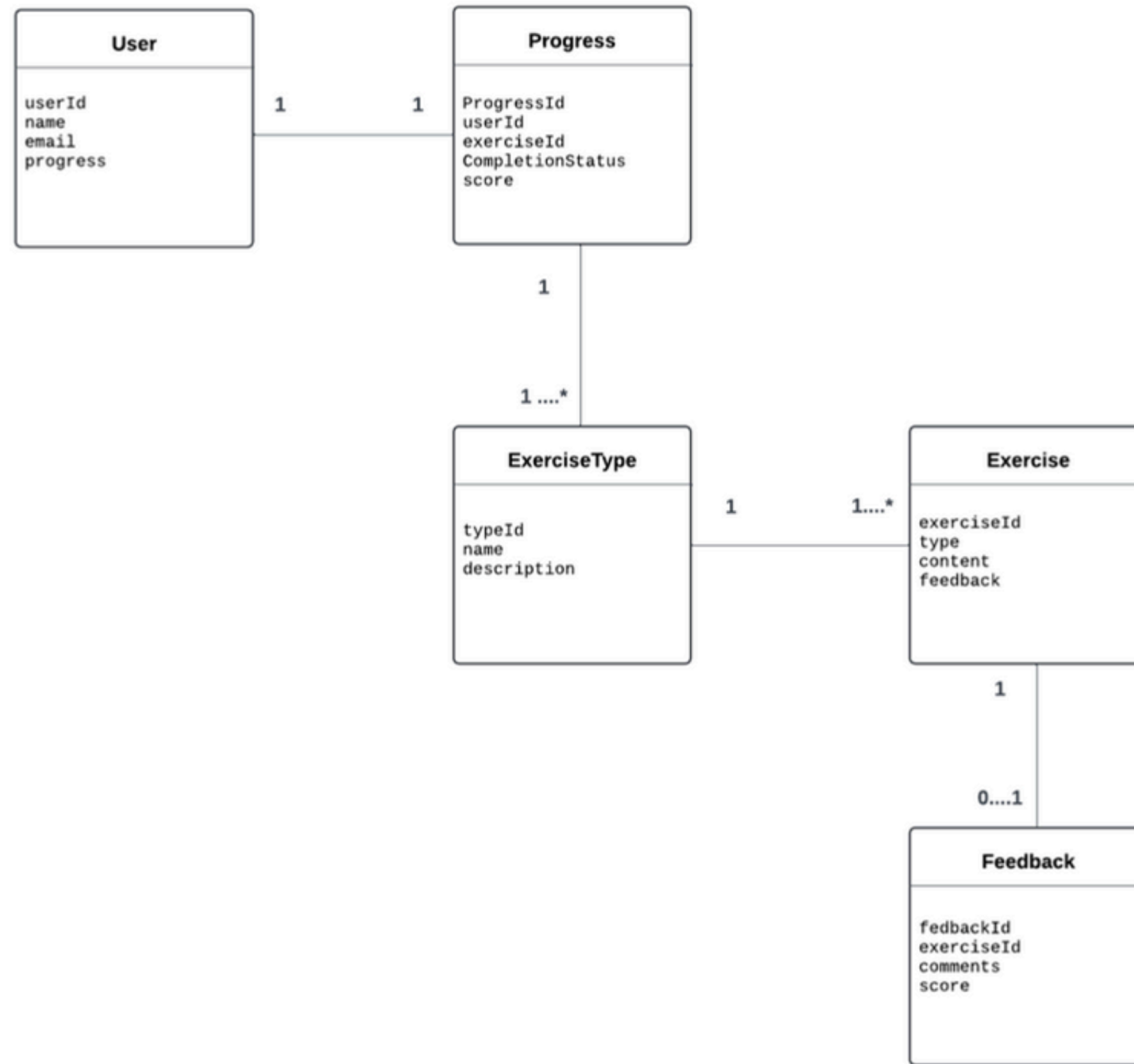
# ACTIVITY DIAGRAM



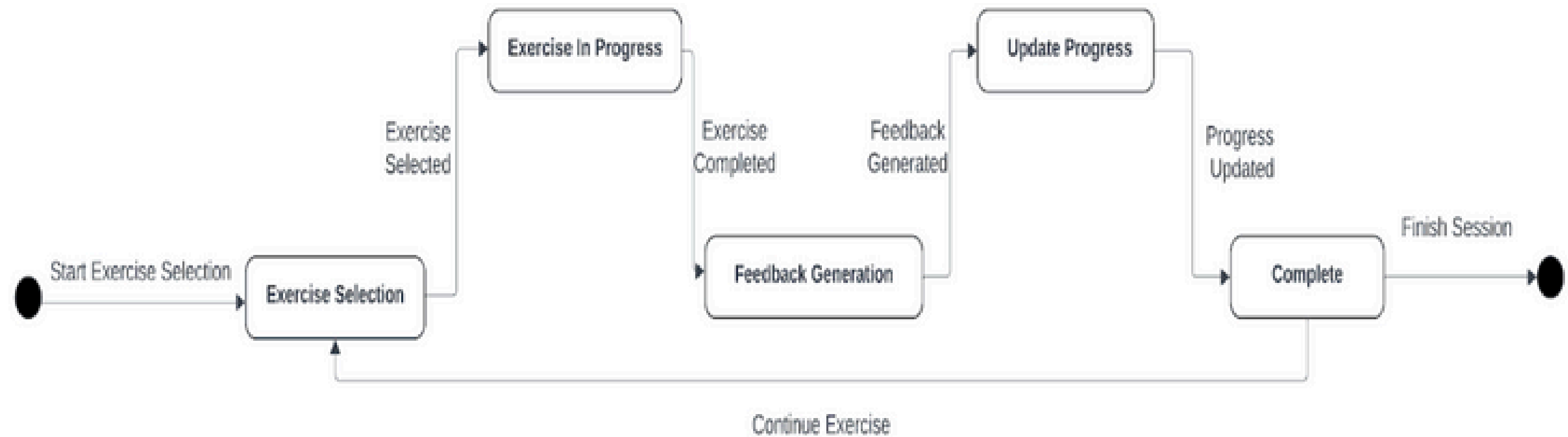
# SYSTEM SEQUENCE DIAGRAM



# DOMAIN MODEL CLASS DIAGRAM



# STATE MACHINE DIAGRAM





# PART 3

## COST BENEFIT ANALYSIS CHART

### Assumptions

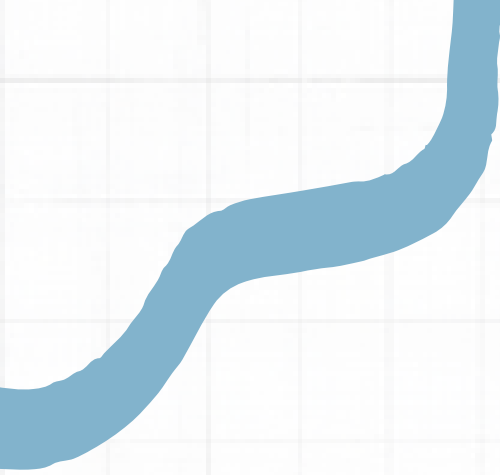
#### I. Costs:

- Initial Development (Year 0): \$500,000
- Operation & Maintenance:
- Year 1: \$50,000
- Increases by 10% annually
- 

#### II. Benefits:

1. Year 1:
  - Users: 50,000
  - Premium Conversion: 5% at \$49.99/year
  - Ad Revenue: \$0.10/user/year
2. Years 2-5:
  - User Growth: 20% annually
  - Premium Conversion: 7%
  - Subscription Price: \$49.99
  - Ad Revenue/user: +10% annually

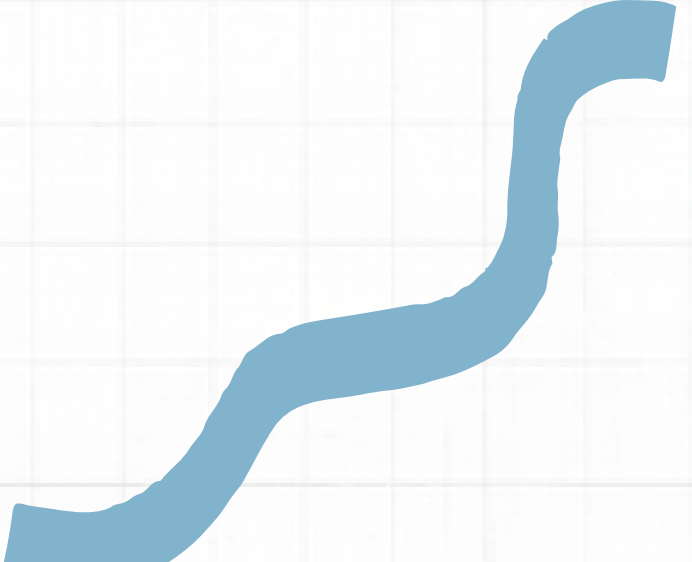
| Cost Benefit Analysis            |            |                                              |           |           |           |           |
|----------------------------------|------------|----------------------------------------------|-----------|-----------|-----------|-----------|
|                                  | Year 0     | Year 1                                       | Year 2    | Year 3    | Year 4    | Year 5    |
| Cost                             |            |                                              |           |           |           |           |
| Development Cost                 | (500,000)  |                                              |           |           |           |           |
| Operating Cost                   |            | (50,000)                                     | (55,000)  | (60,500)  | (66,550)  | (73,205)  |
| Total Cost                       | (500,000)  | (50,000)                                     | (55,000)  | (60,500)  | (66,550)  | (73,205)  |
| Discount Factor (10%)            | 1.00       | 0.9091                                       | 0.8264    | 0.7513    | 0.683     | 0.6209    |
| PV of Cost                       | (500,000)  | (45,454)                                     | (45,454)  | (45,454)  | (45,454)  | (45,454)  |
| Cummulative PV of Costs          | (500,000)  | (545,454)                                    | (590,908) | (636,362) | (681,816) | (727,270) |
| Benefit                          |            |                                              |           |           |           |           |
| Tangible Benefits                | 0          | 129,975                                      | 216,558   | 260,662   | 313,839   | 377,984   |
| Total Benefits                   |            | 129,975                                      | 216,558   | 260,662   | 313,839   | 377,984   |
| Discount Factor (10%)            | 1.00       | 0.9091                                       | 0.8264    | 0.7513    | 0.683     | 0.6209    |
| PV of Benefits                   |            | 117,977                                      | 178,974   | 195,839   | 214,356   | 234,698   |
| Cummulative PV of Benefits       |            | 117,977                                      | 296,951   | 492,790   | 707,146   | 941,844   |
| Cummulative PV of Benefit + Cost | (500,000)  | (427,477)                                    | (293,957) | (143,572) | 25,330    | 214,574   |
| Payback Period                   | 3.96 years | i.e 3 Years 351 Days (0.96 * 365) = 351 days |           |           |           |           |
| Net Present Value (NPV)          | (500,000)  | (427,477)                                    | (293,957) | (143,572) | 25,330    | 214,574   |
|                                  | (500000)   | 79975                                        | 161558    | 200162    | 247289    | 304779    |
| IRR                              | 23%        |                                              |           |           |           |           |
| Profitability Index              | 1.89%      |                                              |           |           |           |           |
|                                  |            |                                              |           |           |           |           |



# PROJECT ITERATION SCHEDULE

- **Iteration 1:** Real-Time Language Translation (4 weeks)
- **Iteration 2:** Interpretation in Social Settings (5 weeks)
- **Iteration 3:** Multilingual Social Networking (6 weeks)
- **Iteration 4:** Cultural Sensitivity Training (3 weeks)
- **Iteration 5:** Collaboration and Connectivity (4 weeks)

**Total Time Estimate: 22 weeks**





# WORK BREAKDOWN SCHEDULE

## Project Schedule

### **1. Project Planning (10 days)**

- Define Scope, Identify Stakeholders, Establish Timeline, Allocate Resources, Risk Assessment

### **2. Analysis (15 days)**

- Market Research, Define Personas, Gather Requirements, Prioritize Use Cases

### **3. Design (20 days)**

- Information Architecture, UI Design, System Architecture, Security Design

### **4. Build (60 days)**

- Language Translation, Social Interpretation, Social Networking, Sensitivity Training, Connectivity Modules

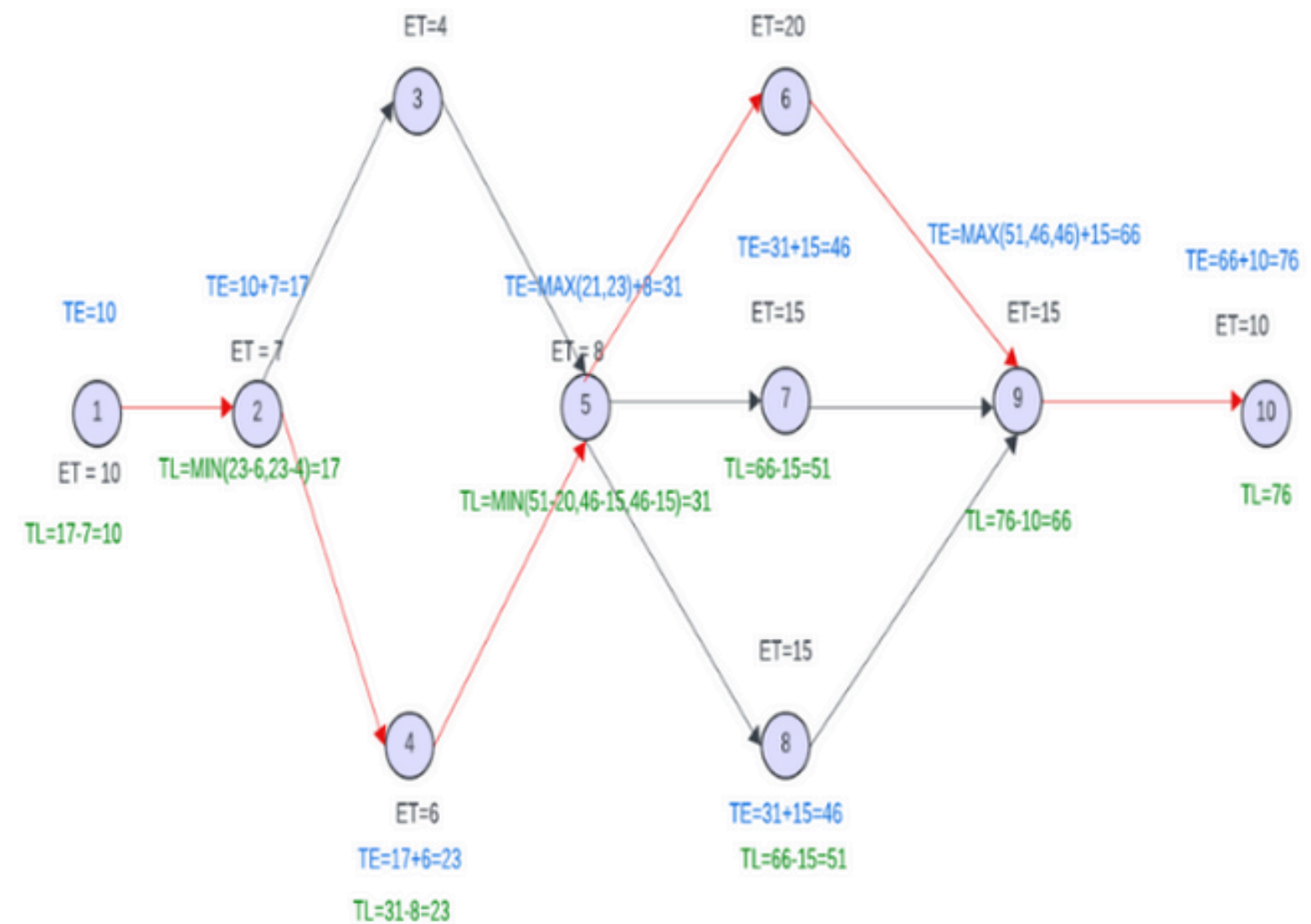
### **5. Testing & Deployment (15 days)**

- Develop Tests, Unit/Integration/UAT, Deploy, Post-Deployment Monitoring

**Total Timeline: 120 days**

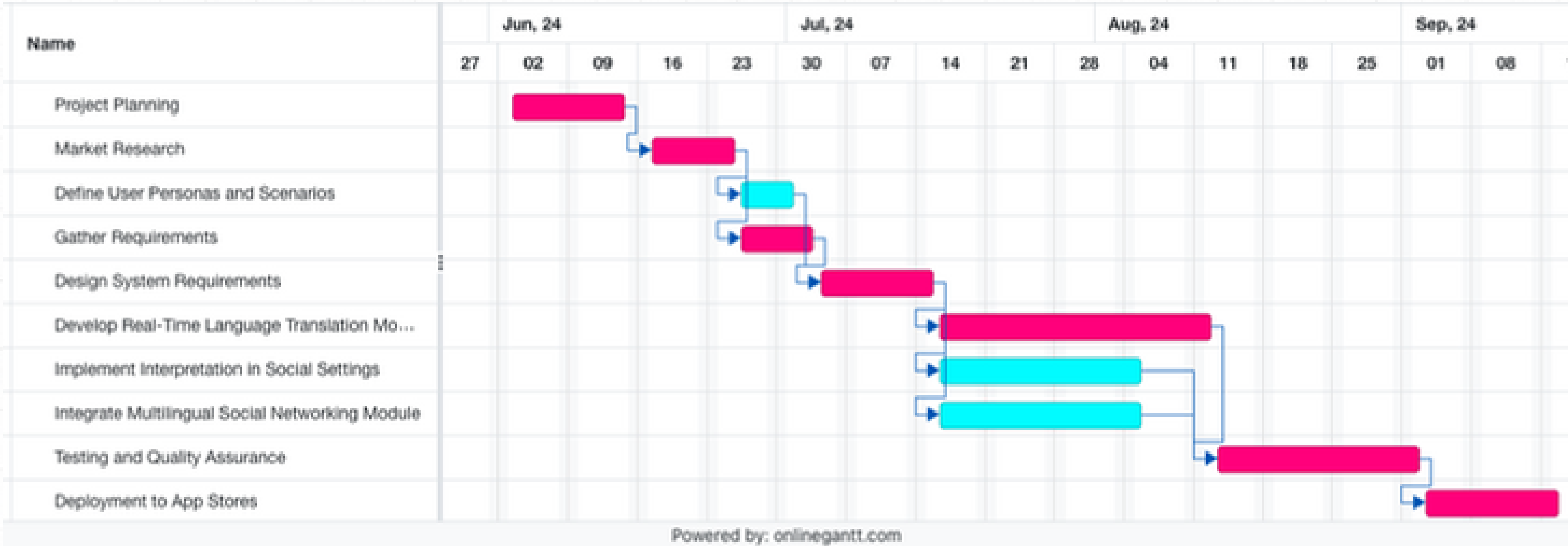
# PROJECT NETWORK DIAGRAM

| Activity                                                  | Preceding Event | Duration | TE | TL | Slack | Critical Path |
|-----------------------------------------------------------|-----------------|----------|----|----|-------|---------------|
| 1. Project Planning                                       | -               | 10       | 10 | 10 | 0     | Yes           |
| 2. Market Research                                        | 1               | 7        | 17 | 17 | 0     | Yes           |
| 3. Define User Personas and Scenarios                     | 2               | 4        | 21 | 23 | 2     | No            |
| 4. Gather Requirements                                    | 2               | 6        | 23 | 23 | 0     | Yes           |
| 5. Design System Requirements                             | 3,4             | 8        | 31 | 31 | 0     | Yes           |
| 6. Develop a Real-Time Language Translation Module        | 5               | 20       | 51 | 51 | 0     | Yes           |
| 7. Implement Interpretation in the Social Settings Module | 5               | 15       | 46 | 51 | 5     | No            |
| 8. Integrate the Multilingual Social Networking Module    | 5               | 15       | 46 | 51 | 5     | No            |
| 9. Testing and Quality Assurance                          | 6,7,8           | 15       | 66 | 66 | 0     | Yes           |
| 10. Deployment to                                         | 9               | 10       | 76 | 76 | 0     | Yes           |



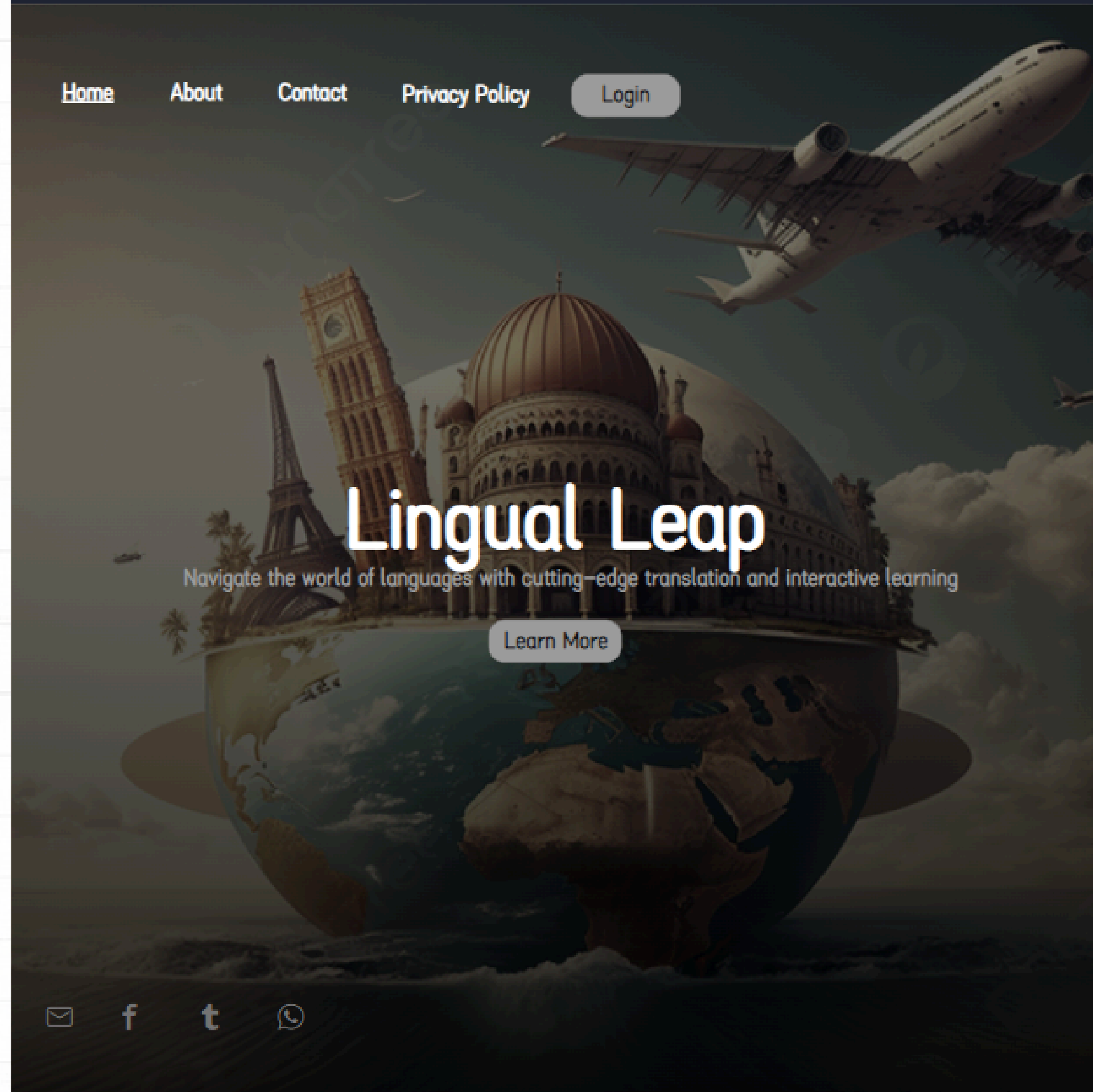


# GANTT CHART





# PART 4



[Home](#)

[Welcome](#)



Let's get started...

Translate Text

Translate Speech

Translate Image

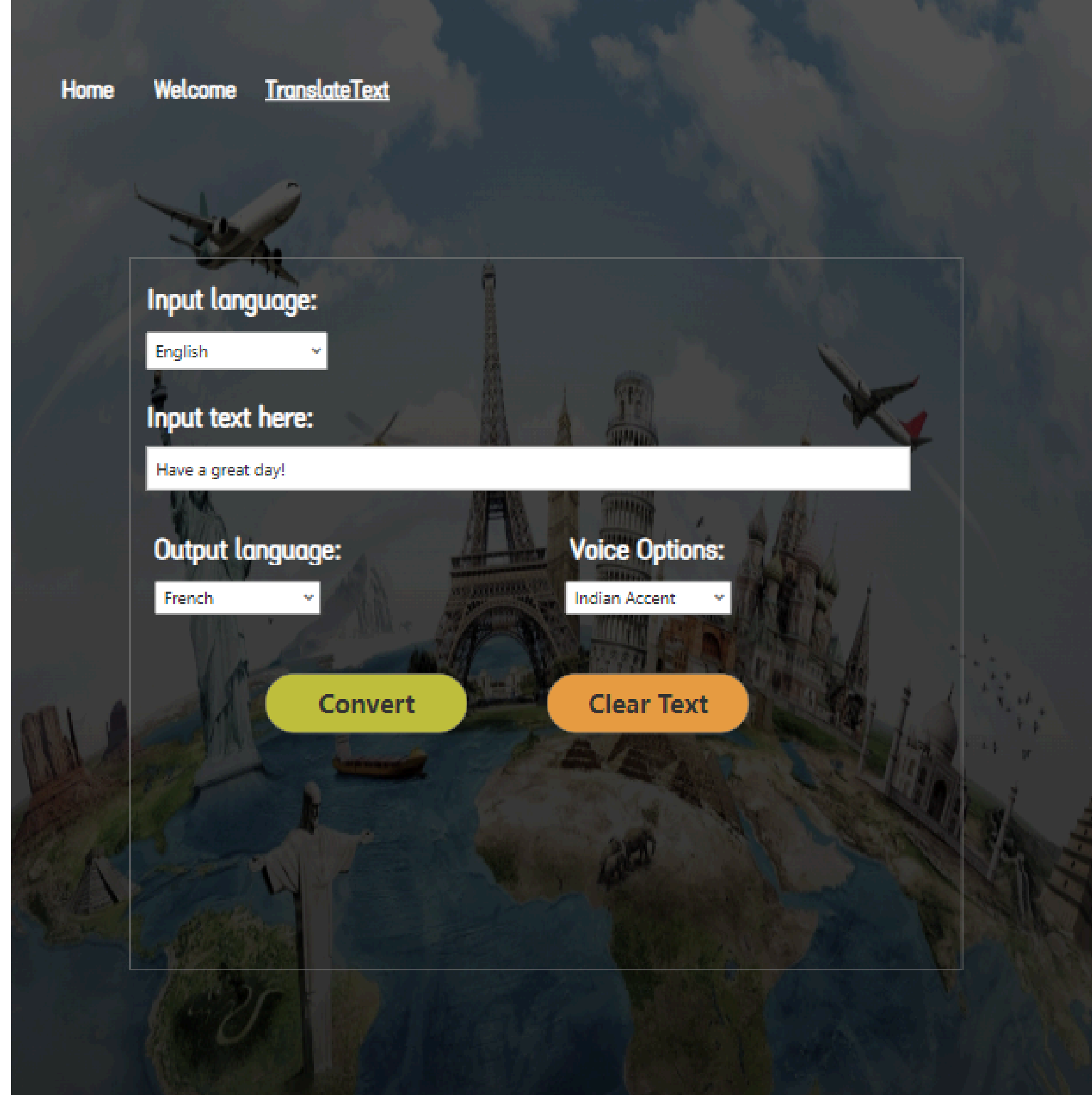
Translate Document

Learn a Language

More..

# USECASE:

## TRANSLATE TEXT



[Home](#) [Welcome](#) [TranslateText](#)

Input language:

English

Input text here:

Have a great day!

Output language:

French

Voice Options:

Indian Accent

Convert

Clear Text

In Progrss...

Input language:

English

Input text here:

Have a Great Day!

Output language:

French

Voice Options:

Indian Accent

Convert

Clear Text

Speech audio:



0:02 / 0:09



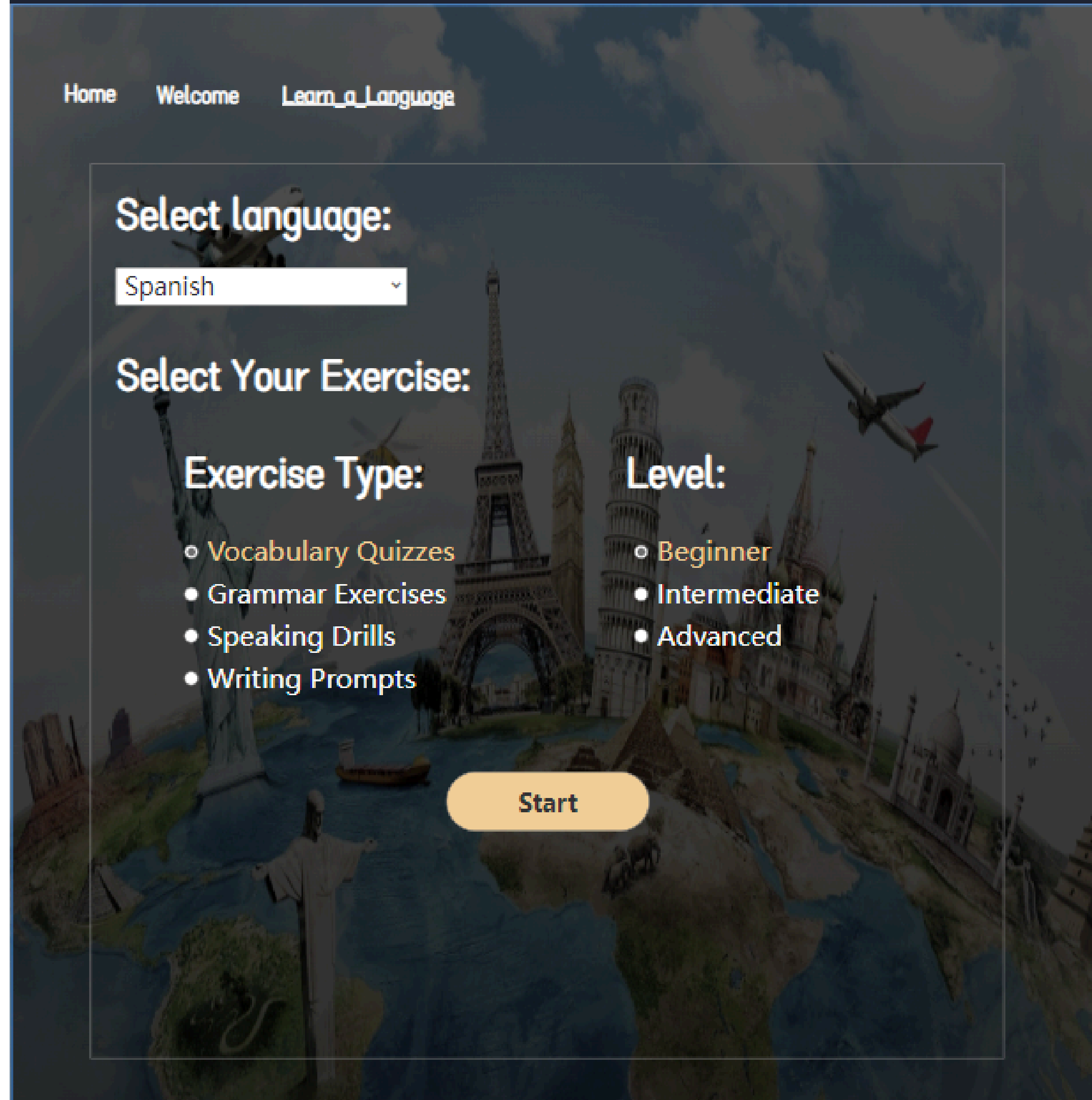
Translated text:

Passez une bonne journée



# USECASE:

## LEARN A LANGUAGE



### Exercise-1:

Choose the correct form of the verb to complete the sentence.

1. Ella \_\_\_\_ a la tienda ayer.

- ☐ va
- ☐ fue
- ☒ ido
- ☐ ir

Submit

### Exercise-1:

Choose the correct form of the verb to complete the sentence.

1. Ella \_\_\_\_ a la tienda ayer.

- ☒ va
- ☒ fue -(correct)
- ☐ ido -(incorrect)
- ☐ ir

Next

Your Progress..



– Good Job!

| Date        | Exercises | Percentage |
|-------------|-----------|------------|
| Jan-05-2024 | Vocablary | 45%        |
| Jan-09-2024 | Writing   | 34%        |
| Jan-23-2024 | Vocablary | 51%        |
| Feb-03-2024 | Grammer   | 67%        |

previous

Overall – Can do Better!  
Areas for improvement

Continue

Exit

THANK YOU

